SoftBuilt PHP Shield

Product Requirements Document (PRD)

A standalone Windows desktop application that imports, protects and deploys PHP projects (starting with EduGate) onto an existing, independently installed XAMPP environment.

Document	Details
Product	SoftBuilt PHP Shield
Target platform	Windows 10 / 11 (64-bit) desktop
First protected project	EduGate (part of the SoftlyBuilt product family)
Version	1.0 (Draft)
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Status	Draft for review

1. Product Overview

SoftBuilt PHP Shield is a standalone Windows desktop executable that lets a developer take a PHP project delivered as a ZIP archive, analyse it, store the original source in an encrypted vault, generate a separate protected production build, and deploy that build to a local XAMPP installation. It manages the deployment, not the web stack: Apache, PHP, MySQL/MariaDB and phpMyAdmin all remain owned by XAMPP.

1.1 Problem statement

PHP applications are delivered as plain-text source. When EduGate is installed on a client machine, anyone with file access can read, copy or modify the code. Developers also lose time configuring Apache virtual hosts, checking PHP extensions and verifying deployments by hand. Shield removes both problems with one tool that protects the code and automates safe deployment.

1.2 Product goals

- Protect source: the original PHP source is stored encrypted and is never placed inside any web-accessible or XAMPP-managed folder.
- Deploy safely: only the files required to run the protected application reach XAMPP, with Apache serving the public directory only.
- Integrate, never replace: detect and use the existing XAMPP; never bundle, fork or install another Apache, PHP or MySQL.
- Verify before claiming success: a deployment is reported successful only after EduGate is confirmed working over HTTP.
- Stay independent: XAMPP and phpMyAdmin remain fully usable on their own, with or without Shield running.
- Run anywhere on Windows: no Python, no extra runtime installs on the target computer.

1.3 Non-goals

- Shipping or installing Apache, PHP, MySQL/MariaDB or phpMyAdmin.
- Replacing or modifying XAMPP Control Panel behaviour.
- Migrating or owning EduGate's database. The database lives in XAMPP's MySQL/MariaDB and is managed through phpMyAdmin.
- Cloud hosting, remote servers, Linux or macOS support (version 1.0).
- Acting as a general PHP IDE or code editor.

2. Critical Architecture Requirement

SoftBuilt PHP Shield is a standalone Windows desktop executable. XAMPP is a completely separate application installed independently on the same computer. The two cooperate through detection, configuration files, Windows services/processes and HTTP checks only. Shield must not embed any XAMPP component.

2.1 Responsibility matrix

Area	SoftBuilt PHP Shield	XAMPP
Project import	Imports PHP projects from ZIP archives	Not involved
Analysis	Analyses PHP code, structure and requirements	Not involved
Source protection	Encrypts and securely stores original source	Never receives original source
Production build	Generates a separately protected production build	Receives only the files required to execute the app

Area	SoftBuilt PHP Shield	XAMPP
Web server	Configures virtual host / document root; never replaces Apache	Provides Apache
PHP	Checks PHP version and required extensions	Provides PHP execution
Database	Detects status; does not own or migrate data	Provides MySQL/MariaDB; stores EduGate's database
Database admin	Offers a shortcut to open phpMyAdmin	Provides phpMyAdmin
Services	Monitors XAMPP services; starts/stops the services it manages	Remains independently startable via XAMPP Control Panel
Status	Displays security and runtime status	Not involved

2.2 Architecture rules (mandatory)

- Do not bundle, replace, fork or install another Apache, PHP or MySQL server.
- Detect and integrate with the existing XAMPP installation.
- Never overwrite existing XAMPP configuration without first creating a verified backup.
- Do not expose protected original source files through Apache or phpMyAdmin.
- The original PHP source stays separate from the executable production deployment.
- The user can open XAMPP independently and use phpMyAdmin to manage the EduGate database at any time.
- Closing Shield stops EduGate's managed runtime only if the user configured it to; XAMPP stays installed and usable either way.
- Shield is a self-contained executable with its own UI, settings, project management and security functions, and requires no Python on the target computer.

3. Users and Use Cases

Persona	Description	Primary needs
Developer / Publisher	Builds EduGate and prepares releases	Import ZIP, protect source, produce builds, deploy for testing
Installer / Technician	Sets up EduGate at a school or client site	Detect XAMPP, one-click deploy, LAN access, verification
Site Administrator	Runs EduGate day to day	See status, start/stop runtime, open phpMyAdmin for the database

Key use cases

- UC-1: Import an EduGate ZIP, review the analysis report and store the project in the encrypted vault.
- UC-2: Generate a protected production build separate from the original source.
- UC-3: Deploy the build to XAMPP, configure Apache, and verify EduGate loads on localhost and LAN.
- UC-4: Monitor Apache and MySQL/MariaDB and see EduGate runtime and security status at a glance.
- UC-5: Roll back to the previous XAMPP configuration or previous deployment after a failure.
- UC-6: Close Shield and, if configured, stop EduGate's managed runtime while XAMPP stays available.

4. Functional Requirements

Priority: **P0** = required for v1.0, **P1** = important, **P2** = later.

4.1 Project import (ZIP)

ID	Requirement	Pri
FR-1.1	Import a PHP project from a .zip archive via file picker or drag and drop.	P0
FR-1.2	Validate the archive before extraction: block path traversal (../), absolute paths, symlinks, and zip bombs (size, file-count and ratio limits).	P0
FR-1.3	Extract to a temporary staging area, never into XAMPP or any web-accessible folder.	P0
FR-1.4	Detect the project root and public directory (for example /public, /public_html or /www) and let the user confirm or change it.	P0
FR-1.5	Support multiple projects, each with name, version, import date and status.	P1
FR-1.6	Support re-import of a new version of an existing project, keeping version history.	P1

4.2 Project analysis

ID	Requirement	Pri
FR-2.1	Scan the project and report: file counts, PHP files, entry points, public vs non-public files, vendor/Composer usage and framework hints.	P0
FR-2.2	Determine the minimum and maximum PHP version the code supports (from composer.json and syntax inspection).	P0
FR-2.3	List required PHP extensions (for example mysqli, pdo_mysql, mbstring, openssl, curl, gd, zip, intl, fileinfo).	P0
FR-2.4	Detect database configuration files, .env files and hard-coded credentials and flag them as warnings.	P0
FR-2.5	Flag risky items: world-writable upload folders, exposed backups (.sql, .bak, .git), debug scripts, phpinfo() calls.	P1
FR-2.6	Produce an exportable analysis report (PDF/HTML).	P2

4.3 Encrypted source vault

ID	Requirement	Pri
FR-3.1	Encrypt every original source file with AES-256-GCM (authenticated encryption) before storing it in the vault.	P0
FR-3.2	Protect vault keys with a user master password (Argon2id key derivation) and Windows DPAPI binding for the local machine/user.	P0
FR-3.3	Store the vault outside XAMPP and outside any Apache-served path (default: %ProgramData%\SoftBuilt\Shield\Vault).	P0
FR-3.4	Check integrity of vault contents (hash/auth tag) on open and report tampering.	P0
FR-3.5	Allow export and restore of an encrypted vault backup protected by a recovery key.	P1
FR-3.6	Auto-lock the vault after a configurable idle period and on application exit.	P1

4.4 Protected production build

ID	Requirement	Pri
FR-4.1	Generate a production build as a separate artefact from the vault; it must never contain the original source layout verbatim if protection is enabled.	P0
FR-4.2	Build contents: only runtime-required files; exclude tests, docs, .git, .env.example, build scripts, backups and other non-runtime files.	P0

ID	Requirement	Pri
FR-4.3	Apply a configurable protection profile: (a) minify/strip comments, (b) identifier obfuscation, (c) optional integration with a user-supplied commercial PHP encoder/loader installed in XAMPP.	P0
FR-4.4	Separate public files (Apache document root) from private application files (outside document root).	P0
FR-4.5	Inject deployment-time configuration (database host, name, user) through a generated config file; database passwords never enter the vault or logs in plain text.	P0
FR-4.6	Create a build manifest listing every file with its SHA-256 hash, and sign the build with a Shield-held key.	P1
FR-4.7	Keep a history of builds and allow rebuilding from the vault.	P1

4.5 XAMPP detection and integration

ID	Requirement	Pri
FR-5.1	Auto-detect the XAMPP installation directory by checking the Windows registry, common paths (C:\xampp and other drives) and running processes; allow manual selection.	P0
FR-5.2	Validate the installation: presence of apache\bin\httpd.exe, php\php.exe, mysql\bin, phpMyAdmin and xampp-control.exe.	P0
FR-5.3	Detect the installed PHP version by running the XAMPP php.exe -v and -m; list loaded extensions.	P0
FR-5.4	Compare PHP version and extensions with the project requirements and present a clear pass / warn / fail checklist.	P0
FR-5.5	Detect Apache and MySQL/MariaDB status (running or stopped, service or process, ports in use, conflicts with IIS/Skype/other software).	P0
FR-5.6	Detect phpMyAdmin availability and URL.	P1
FR-5.7	Show detected XAMPP data (path, versions, ports) in a read-only summary; re-detect on demand.	P0
FR-5.8	If XAMPP is not found, guide the user to install it independently; Shield must never install or download XAMPP itself.	P0

4.6 Apache and deployment configuration

ID	Requirement	Pri
FR-6.1	Create a dedicated EduGate deployment directory (default: <xampp>\softbuilt-apps\edugate\), separate from htdocs.	P0
FR-6.2	Configure Apache to serve EduGate's public directory only, using a dedicated virtual host or alias file included from httpd.conf.	P0
FR-6.3	Before touching any XAMPP file (httpd.conf, extra\httpd-vhosts.conf, hosts, php.ini), create a timestamped backup and verify it; never overwrite without a backup.	P0
FR-6.4	Keep Shield's changes minimal and isolated: prefer one added include file (for example conf\extra\softbuilt-shield.conf) over editing existing directives.	P0
FR-6.5	Validate the new configuration with httpd.exe -t before applying; revert automatically if invalid.	P0
FR-6.6	Configure local access (localhost / local hostname such as edugate.local) and optional LAN access with Require ip ranges and a Windows Firewall rule, shown to the user for approval.	P0
FR-6.7	Enable only the Apache modules and PHP extensions required, with user approval, and back up php.ini before any change.	P1
FR-6.8	Block access to sensitive paths: dotfiles, .env, vendor internals, storage/, logs and backups.	P0
FR-6.9	Ensure phpMyAdmin cannot browse or expose the vault, source folders or production private files.	P0

4.7 Deployment and verification

ID	Requirement	Pri
FR-7.1	Copy only the production build files to the EduGate deployment directory using atomic release folders so a failed deployment never leaves a half-updated site.	P0
FR-7.2	Run a pre-flight checklist: XAMPP found, PHP compatible, ports free, config backed up, disk space, permissions.	P0
FR-7.3	Restart or gracefully reload Apache as required, with user confirmation.	P0
FR-7.4	Post-deployment verification: HTTP check against localhost (and LAN address if enabled), expect a healthy EduGate response, confirm no PHP errors, and confirm a database connection test.	P0
FR-7.5	Confirm negative checks: requests for private files, .env, the vault path and source file extensions return 403/404.	P0
FR-7.6	Report deployment success only after all required verification checks pass; otherwise report failure with exact reasons.	P0
FR-7.7	One-click rollback to the previous release and previous XAMPP configuration backup.	P0
FR-7.8	Keep a deployment history with date, build hash, result and verification details.	P1

4.8 Service monitoring and runtime control

ID	Requirement	Pri
FR-8.1	Display live status of Apache, MySQL/MariaDB and the EduGate site (running, stopped, error), refreshed every few seconds.	P0
FR-8.2	Start and stop only the services that Shield itself started or manages for EduGate; the user can still control XAMPP through its own Control Panel.	P0
FR-8.3	Request administrator elevation only when required (service control, firewall rule, protected folders) and explain why.	P0
FR-8.4	Setting: 'Stop EduGate runtime when Shield closes' (default off). When off, closing Shield leaves services running. When on, stop only what Shield started. XAMPP is never uninstalled or altered by closing Shield.	P0
FR-8.5	Detect services stopped or started outside Shield (for example via XAMPP Control Panel) and update status without error.	P0
FR-8.6	Buttons to open EduGate in the browser, open phpMyAdmin, and open XAMPP Control Panel.	P1
FR-8.7	Notify the user if a managed service crashes, with a link to the relevant log.	P1

4.9 Security and runtime status dashboard

ID	Requirement	Pri
FR-9.1	Single dashboard showing: XAMPP detected, PHP version/extension compatibility, Apache and database state, vault locked/unlocked, last build, last deployment, and last verification result.	P0
FR-9.2	Security checklist with pass/warn/fail: source not in web path, only public directory served, config backed up, sensitive paths blocked, debug mode off, LAN exposure state.	P0
FR-9.3	Show an overall protection score or status badge and a prioritised list of fixes.	P1
FR-9.4	Tamper check: compare deployed files with the signed build manifest and alert on changes.	P1

4.10 Settings, logging and audit

ID	Requirement	Pri
FR-10.1	Settings: XAMPP path, vault location, deployment directory, hostname and port, LAN access, auto-lock, stop-on-exit, theme, update checks.	P0
FR-10.2	Local audit log of imports, builds, deployments, config changes, logins/unlocks and rollbacks, without secrets.	P0
FR-10.3	Export logs for support, with secrets automatically redacted.	P1

5. User Interface

Shield has its own native Windows user interface; it does not run inside XAMPP or a browser tab that depends on Apache.

Screen	Purpose	Key elements
Dashboard	Overall state	XAMPP, Apache, DB, EduGate status cards; security badge; quick actions
Projects	Project list	Add ZIP, name, version, status, open details
Project Detail / Analysis	Review analysis	PHP/extension requirements, warnings, public directory selection
Vault	Source protection	Lock state, integrity check, backup/restore
Builds	Production builds	Protection profile, build list, manifest, hashes
Deploy	Guided deployment	Pre-flight checklist, config preview/diff, deploy, verification results, rollback
XAMPP Integration	Environment	Detected path, versions, ports, extensions, backups, re-detect
Services	Runtime control	Apache/DB start-stop (managed), logs, open phpMyAdmin
Security	Hardening checklist	Pass/warn/fail items and recommended fixes
Settings & Logs	Preferences, audit	All settings, log viewer, export

UX principles

- Guided, wizard-style deployment with plain-language explanations and a config diff before any change is applied.
- Destructive or system-level actions always require explicit confirmation.
- Clear separation of 'Shield-managed' and 'XAMPP-owned' items in the UI so users understand what Shield will and will not touch.
- Non-technical installers can complete a deployment without editing a file by hand.

6. Deployment Workflow

Step	Action	Result
1	Detect XAMPP and read PHP/Apache/DB status	Environment summary
2	Import and validate the project ZIP	Safe staging copy
3	Analyse project	Requirements and warnings
4	Encrypt and store original source in the vault	Source secured outside XAMPP

Step	Action	Result
5	Generate protected production build	Signed build with manifest
6	Pre-flight checks and compatibility verdict	Pass / warn / fail
7	Back up affected XAMPP configuration files	Verified backups
8	Copy build to dedicated deployment directory (new release folder)	Files in place
9	Write isolated Apache config, test with httpd -t, reload Apache	Public directory served only
10	Verify over HTTP (localhost/LAN), DB connection, negative path checks	All checks green
11	Switch release live and report success; otherwise roll back automatically	Deployment recorded

7. Folder and Data Layout

Location	Owner	Contents	Web-accessible?
%ProgramData%\SoftBuilt\Shield\Vault	Shield	Encrypted original source, key material	No
%ProgramData%\SoftBuilt\Shield\Builds	Shield	Protected production builds and manifests	No
%ProgramData%\SoftBuilt\Shield\Backups	Shield	Backups of XAMPP config files, vault backups	No
%AppData%\SoftBuilt\Shield	Shield	Settings, logs, project metadata	No
<xampp>\softbuilt-apps\edugate\releases\<id>\app	Shield deploys, XAMPP executes	Private runtime PHP files	No (outside document root)
<xampp>\softbuilt-apps\edugate\current\public	Shield deploys, XAMPP serves	Public entry point and assets only	Yes (only this)
<xampp>\mysql\data	XAMPP	EduGate database	No
<xampp>\htdocs, phpMyAdmin	XAMPP	Untouched by Shield	As configured by user

8. Security Requirements

8.1 Threat model

Threat	Mitigation
Client or visitor reads PHP source	Original source only in encrypted vault; production build protected and private files outside document root
Source exposed through web server or phpMyAdmin	Only public directory served; deny rules for sensitive paths; vault never under XAMPP; negative HTTP tests
Malicious ZIP (zip-slip, bombs, symlinks)	Strict archive validation and staging before any use
Config corruption of XAMPP	Backup before change, isolated include file, httpd -t validation, automatic rollback
Stolen vault files	AES-256-GCM, Argon2id master password, DPAPI binding, integrity checks
Tampering with deployed files	Signed manifest and hash comparison

Threat	Mitigation
Unintended LAN exposure	LAN off by default; explicit approval; IP-range restriction; visible status
Secrets leaking into logs	Redaction and no plain-text credential storage

8.2 Honest limitation

PHP is an interpreted language, and XAMPP's PHP must be able to read whatever it executes. Obfuscation raises the effort needed to copy or reverse code but cannot make it impossible. Stronger protection requires a commercial PHP encoder with a loader extension installed into XAMPP (supported as an optional integration, FR-4.3c). The product promise is therefore: original source is never exposed, and the deployed code is hardened, not that deployed code is unbreakable. This should be stated plainly in the UI and documentation.

9. Non-Functional Requirements

Category	Requirement
Packaging	Single standalone Windows executable (self-contained, installer optional). No Python, no .NET runtime install required on the target PC.
Platform	Windows 10 (1809+) and Windows 11, x64.
Performance	Start in under 3 seconds; analyse a 50 MB project in under 30 seconds; status refresh every 3-5 seconds without UI freezing.
Reliability	Every system change is reversible; no partial deployments; safe if killed mid-operation.
Compatibility	Work with common XAMPP versions (PHP 8.x, MariaDB/MySQL) and non-default install paths and ports.
Privacy	Fully offline operation; no telemetry by default; update checks opt-in.
Usability	Install-and-deploy by a non-expert within 10 minutes using the guided flow.
Accessibility	Keyboard navigation, readable contrast, scalable fonts.
Localisation	English first; text externalised for later translation.

10. Recommended Technical Approach

The requirement that Shield runs without Python on the target machine, combined with the desktop stack preferences from earlier SoftlyBuilt work, points to C# / .NET with a self-contained single-file publish.

Concern	Recommendation
UI	WPF or WinUI 3 (native, no browser dependency); Electron is a viable alternative if the team prefers web UI skills
Packaging	dotnet publish: self-contained, win-x64, single-file; optional MSIX / Inno Setup installer
Cryptography	AES-256-GCM via .NET cryptography APIs, Argon2id via a vetted library, DPAPI (ProtectedData)
Local storage	SQLite for project metadata and logs (inside Shield's own data folder; unrelated to XAMPP's database)
XAMPP integration	Registry/path discovery, process execution of php.exe and httpd.exe, Windows Service Control Manager, config file parsing and writing with backups
Verification	Built-in HTTP client for health and negative-path checks
Code analysis	Bundled PHP parser library inside Shield (not the XAMPP PHP) for static analysis; XAMPP's php.exe used only for version/extension queries

11. Acceptance Criteria

- AC-1: On a PC with XAMPP installed in a non-default folder, Shield detects it automatically and reports PHP version, extensions, and Apache/DB status correctly.
- AC-2: Shield launches and runs on a PC with no Python installed.
- AC-3: Importing a malicious ZIP (path traversal) is blocked with a clear message and no files written outside staging.
- AC-4: After import, original source exists only as encrypted data in the vault; a search of XAMPP folders finds no original source.
- AC-5: Deployment backs up all modified XAMPP files first and the backups can be restored with one click.
- AC-6: Apache serves only EduGate's public directory; requests for private files, .env and vault paths return 403/404.
- AC-7: EduGate loads on localhost, and on LAN when enabled, and Shield shows success only after verification passes.
- AC-8: A deliberately broken deployment (for example missing PHP extension) is reported as failed and automatically rolled back.
- AC-9: The user can open XAMPP Control Panel and phpMyAdmin independently and manage the EduGate database, with Shield open or closed.
- AC-10: With 'stop on exit' off, closing Shield leaves EduGate running; with it on, only Shield-started services stop and XAMPP remains installed and usable.
- AC-11: Shield never installs, bundles or launches its own Apache, PHP or MySQL.

12. Milestones

Phase	Scope	Outcome
M1 - Foundation	App shell, settings, logging, XAMPP detection and status	Detects and monitors XAMPP
M2 - Import and analysis	ZIP validation, project list, analysis report	Projects imported and analysed
M3 - Vault	Encryption, master password, integrity, backup	Source secured
M4 - Build	Protection profiles, build separation, manifest	Protected production builds
M5 - Deploy	Config backup, Apache include, deploy, verify, rollback	End-to-end EduGate deployment
M6 - Hardening and release	Security dashboard, LAN mode, installer, testing on multiple XAMPP versions	v1.0 release

13. Risks, Assumptions and Open Questions

Risks

Risk	Impact	Mitigation
Obfuscation can be reversed	Source partially recoverable	Be transparent; offer commercial encoder integration
XAMPP versions and paths vary	Detection failures	Multiple detection methods plus manual path
Port conflicts (80/443/3306)	Apache or DB will not start	Conflict detection and guided alternate ports
Antivirus flags a protector/obfuscator tool	Distribution issues	Code signing certificate; vendor allow-listing

Risk	Impact	Mitigation
Admin rights needed for services/firewall	Blocked actions	Targeted elevation with explanation; degrade gracefully
Client edits XAMPP config manually	Drift	Re-verify on launch and before deploy

Assumptions

- XAMPP for Windows is installed by the user before using Shield.
- EduGate is a PHP + MySQL/MariaDB application with an identifiable public entry directory.
- Windows only for v1.0.

Open questions

- Which protection level is required for v1.0: obfuscation only, or also an optional commercial encoder integration?
- Is licensing or per-machine binding of the protected build required?
- Will Shield be sold or distributed to clients, or used internally by SoftlyBuilt only?
- Should Shield support other PHP projects beyond EduGate in v1.0, or EduGate-specific profiles only?
- Is HTTPS (local certificate) required for LAN access in v1.0?